Constantine 1-Frères Mentouri University
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LMD L1 SM **2026-2027**

PRACTICAL WORK (TP) — SERIES No. 2 Getting Started with Python

Module: Informatique 1 **Level:** L1 Common Core — Sciences de la Matière **Duration:** 2 sessions × 1 h 30 **Tools:** Python 3 + IDLE ·

Student version — Chapter 2 « Débuter avec Python ». Write the algorithm on paper, write the program yourself, then compare your results with the expected ones. ★ = bonus for fast students (not compulsory). Tick ☐ when the exercise is done **and tested**.

METHOD (every exercise)

1. Read twice: identify **data**, **results**, **formula**.
2. Write the **algorithm** (START – READ – COMPUTE – DISPLAY – END).
3. IDLE: *File → New File*, code, save as `tp2_ex1.py`, `tp2_ex2.py`… then **F5**.
4. Test with 2 data sets, one being a limit case (0, negative, decimal).

BY THE END YOU CAN…

write, save and run a Python script · use `print()` and `input()` · create variables and constants and check types with `type()` · convert with `int()`, `float()` · use `+ - * / // % **` · turn a scientific formula into Python (with units) · read an error message.
Good practice: meaningful names, a comment line at the top of each file, 4-space indentation, **units** in every prompt: `"Volume V (L): "`.

QUICK REFERENCE (memento)

Comment	`# this line is ignored`	everything after `#` is ignored
Display	`print("A =", x)` · `print(f"A = {x:.2f} cm²")`	`sep=` and `end=` change separator / ending; `{x:.2f}` = 2 decimals
Input	`x = float(input("x (m): "))`	`input()` **always returns a str** → convert with `int()` / `float()`
Assignment, constant	`age = 19` · `PI = 3.14159`	`=` stores a value (not a maths equality); CAPITALS = constant by convention
Module	`import math` → `math.sqrt(x)`, `math.pi`	import before use, otherwise NameError
Types	`int` 25 · `float` 3.14 (dot!) · `str` "25" · `bool` True/False · `complex` 2+3j	`type(x)` shows the type; `int(3.99)` = 3 (truncates); `round(x, n)`
Operators	`+ - * / // % **` → `7/2`=3.5 · `7//2`=3 · `7%2`=1 · `2**3`=8	Priority: `( )` → `**` → `* / // %` → `+ -` ; `2 + 3 * 4 ** 2` = 50
Valid names	`my_age`, `V_mL`, `total2`	letters, digits, `_` · not starting with a digit · no spaces · not a keyword (`class`, `if`…) · `Age` ≠ `age`

SESSION 1 — Input / output, variables and data types (Exercises 1–5 · 90 min)

Exercise 1 — Hello, World! (10 min) ☐

Start your file with two comment lines (exercise, your name, date), then display **Hello, World!**. Next: (a) show it on **two lines** with a single `print` (hint: `\n`); (b) show the date 2026-10-07 using `sep="-"`.

Test → `Hello, World!` · `Hello,` / `World!` · `2026-10-07`

Exercise 2 — Sum of two integers (15 min) ☐

Read two **integers** `a` and `b` typed by the user, compute their sum and display it.

Test → (12, 30) → `42` · (−7, 4) → `-3`

★ **Bonus:** Also display the difference, product, integer quotient `//` and remainder `%`. Test (17, 5) → `22, 12, 85, 3, 2.`

Exercise 3 — Real numbers (15 min) ☐

Read two **real numbers** `x1` and `x2` (both non-zero). Display their sum `som`, their product `prod` and their quotient `div = x1 / x2`.

Test → (7.5, 2.5) → `10.0` · `18.75` · `3.0`

★ **Bonus:** (a) Display the quotient and the mean (x1 + x2) / 2 with **2 decimals** (f-string); test (1, 3) → `0.33` and `2.00`. (b) Run with x2 = 0 and read the error message.

Exercise 4 — Name and age (10 min) ☐

Ask the user for their **name** and **age**, then display them in two short sentences.

Test → Amina, 19 → `Hello, Amina` / `You are 19 years old.`

★ **Bonus:** Use **one f-string** and compute the approximate year of birth with the constant `CURRENT_YEAR = 2026` → `2007`.

Exercise 5 — Variables and data types (20 min) ☐

Create five variables: `name` (text), `age` (integer), `length` (height in m, real), `is_student` (boolean) and `z` (complex, e.g. 2+3j). Display each **value and its type** with `type()`. Then **swap** `a = 3` and `b = 7` (i) with a temporary variable, (ii) with `a, b = b, a`.

Test → `Amina <class 'str'>` … `(2+3j) <class 'complex'>` · after the swap: a = 7, b = 3

★ **Bonus:** Predict, then check: `int("25") + 1`, `float(3)`, `int(3.99)`, `bool(0)`, `5 / 2`, `5 // 2` → `26, 3.0, 3, False, 2.5, 2.`

Session 1 wrap-up: ☐ I can write, save and run a `.py` file ☐ `input()` returns a string and I convert it ☐ I use `print` with `sep`, `end`, f-strings ☐ I can name the five types.

SESSION 2 — Scientific calculations (Exercises 6–9 · 90 min)

Exercise 6 — Area of a circle (constants) (15 min) ☐

Define the constant `PI = 3.14159` and the radius `R = 1.5` (cm). Compute and display the area A = PI × R².

Test → `7.0685775 cm²` → rounded to 2 decimals: `7.07`

★ **Bonus:** Add the circumference 2·PI·R and the volume of a sphere 4/3·PI·R³ (`import math`: compare with `math.pi`) → `9.42, 14.14`.

Exercise 7 — Area of a triangle (Heron) (25 min) ☐

Read the sides `a`, `b`, `c` (same unit). Compute p = (a+b+c)/2 then area = √(p (p−a)(p−b)(p−c)) using `math.sqrt`. Display p and the area rounded to 2 decimals.

Test → (3, 4, 5) → `p = 6.0`, area `6.0` · (7, 8, 9) → `p = 12.0`, area `26.83`

★ **Bonus:** For (3, 4, 5): inscribed radius r = area / p and height h = 2·area / c → `1.0, 2.4`. ★★ Challenge: try (1, 2, 5): which error? Why?

Exercise 8 — Moles and molar concentration (20 min) ☐

A mass `m` (g) of solute with molar mass `M` (g/mol) is dissolved to make a volume `V` (L). Read `m`, `M`, `V`, then compute and display n = m / M (mol) and C = n / V (mol/L).

Test → 5.85 g NaCl, M = 58.5 g/mol, V = 0.5 L → `n = 0.1 mol`, `C = 0.2 mol/L`

★ **Bonus:** (a) Mass of NaOH (M = 40) for 250 mL at 0.1 mol/L (m = C·V·M) → `1.0 g`. (b) Volume of a 1 mol/L stock for 100 mL at 0.1 mol/L (C1·V1 = C2·V2) → `10 mL`.

Exercise 9 — Synthesis — laboratory report (20 min) ☐

Read the **name** of a substance, its mass `m` (g), molar mass `M` (g/mol) and volume `V_mL` in **millilitres**. Convert the volume to litres, compute n and C and display a neat report: m and M with 2 decimals, V with 3, n and C with 4.

Test → NaCl, 5.85, 58.5, 500 → `V = 0.500 L`, `n = 0.1000 mol`, `C = 0.2000 mol/L`

★ **Bonus:** Add a line with the concentration in mmol/L (C × 1000) → `200.0`.

Session 2 wrap-up: ☐ I use `math.sqrt` and constants ☐ I check **units** and **operator priority** ☐ I round and format results ☐ I verified one result by hand.

CHECK YOUR UNDERSTANDING — predict the output WITHOUT running, then check in IDLE (homework)

1	`print(5 // 2, 7 % 3, 2 ** 3, 7 / 2)`		**5**	`x = 10 ; x = x + 5 ; print(x)`	
2	`print("3" + "4", "3" * 3)`		**6**	`print(round(3.14159, 2), abs(-4), max(2, 9, 4))`	
3	`print(int("7") + 1)`		**7**	`print(10 / 2)`	
4	`print(type(True), type(3.0), type("3"))`		**8**	`print("a", "b", sep="-")`	

COMMON ERROR MESSAGES — read the LAST line of the message and the line number

Message	Example	Probable cause	How to fix
NameError: name 'Print' is not defined	`Print("hi")`	wrong case, misspelled name, variable not defined yet, missing quotes or import	check spelling; define before use; add quotes / `import`
SyntaxError: unterminated string literal	`print("hi)`	missing quote or parenthesis, misuse of symbols	close the quote; read the line shown by IDLE
IndentationError: unexpected indent	line starting with spaces	spaces at the start of a line that must not be indented	remove the spaces
ValueError: invalid literal for int()	`int("abc")`, `int("3.5")`	text that cannot be converted to a number	type a valid number; use `float` for decimals
TypeError: can only concatenate str (not "int") to str	`"a" + 1`	mixing text and number	`int()` / `str()` or use commas in `print`
ZeroDivisionError: division by zero	`5 / 0`	denominator equal to zero	check the data before dividing
ValueError: math domain error	`math.sqrt(-24)`	square root of a negative number	check the data (e.g. triangle inequality)

Assessment (/10): correct results 8 · right concepts (conversions, operators, units) 4 · code quality (names, comments, indentation) 4 · tests and hand check 2 · autonomy and punctuality 2.